



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Generative AI for Image and Video Creation

TGS Ref No: TGS-2020505925

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 11.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
10.0	Legacy release (superseded)	Legacy release delivered as 'Image and Video Processing with OpenCV' — 182 slides, six OpenCV topics, Raspberry Pi setup and Darknet/YOLO installation sequences. Retained as the content-coverage reference for this revision.	Dr. Alfred Ang
11.0	6 September 2026	Rebuilt for the registered title 'Generative AI for Image and Video Creation'. Content restructured onto the five topics published on the course page while preserving all six approved Learning Units, ELO1-ELO6, K1-K13 and A1-A8. New 204-slide deck in the current all-white house design with 18 native, editable PowerPoint charts, every figure labelled MEASURED / MODELLED / PUBLISHED / ILLUSTRATIVE. Twelve new self-contained labs replace the legacy activity list. Legacy Raspberry Pi provisioning and Darknet build sequences removed; detection now uses the two machine-learning detectors that ship inside opencv-python, so no weights are downloaded. Vendor model identifiers verified against official documentation on 6 September 2026, including the deprecation of the imagen-4.0-* and veo-3.0-generate-001 identifiers. Assessment instruments, question and task counts, ordering, K/A mapping and timings preserved exactly from the registered papers (WA 13 questions / 60 min; PP 5 tasks / 90 min).	Dr. Alfred Ang

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Course Information

Course Title	Generative AI for Image and Video Creation
Registered Public Title	WSQ - Generative AI for Image and Video Creation
WSQ Course Reference	TGS-2020505925
Skills Framework TSC	Computer Vision Technology (ICT-DIT-4022-1.1), Proficiency Level 4
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 16 programme hours (7.5 h classroom facilitation + 6 h practical + 2.5 h assessment)
Daily Timing	Day 1 9:00 am–6:30 pm; Day 2 9:00 am–6:40 pm. Administration from 8:45 am; 1-hour lunch and two 15-minute tea breaks are outside programme hours.
Mode	Instructor-led classroom facilitation with hands-on practical work; 12 self-contained labs run locally with no paid service
Trainer	Dr. Alfred Ang
Trainer-to-learner ratio	1:3 to 1:10
Assessor-to-candidate ratio	1:3 to 1:10 for both WA(Q&A) and PP
Slide deck referenced	Generative AI for Image and Video Creation-v11.0.pptx — 204 slides (version v11.0, 6 September 2026)

Programme-Hours Reconciliation and Time Assumptions

Planning timetable: preserve 450 minutes classroom facilitation, 360 minutes practical and 150 minutes assessment over two days, 480 programme minutes per day. Lunch, tea breaks and administration are additional, non-instructional time. Clock times below are a proposed trainer timetable, not a claim about a currently booked session. Each approved LU retains CR75 minutes and PP0/60/60/120/60/60 minutes respectively. Topic4 combines LU4 and LU5 but preserves both budgets. Lab01 is a facilitated needs-analysis discussion within LU1 classroom facilitation. Lab extensions are outside core class time. The public course page says two assessment hours; the authenticated LMS record and supplied proposal/assessment plan specify WA60 + PP90. The verified 150 minutes is used.

Component	Approved (min)	Scheduled (min)	Reconciles
Classroom facilitation	450	450	YES
Practical / practicum	360	360	YES
Assessment	150	150	YES
TOTAL PROGRAMME HOURS	960	960	YES

Note on the published assessment duration: the public course page states a 2-hour assessment. The registered Assessment Plan (TGS-2020505925 V1.0) and the approved Course Proposal (CA-WSQ-2020-013290-v2) both specify WA(Q&A) 60 minutes plus PP 90 minutes = 2.5 hours, and that is what the authenticated LMS record carries. The 2.5-hour figure is authoritative and is used throughout this Lesson Plan.

Learning Outcomes and Competency Mapping

On completion of this course, learners will be able to:

- LO1: Understand basic vision systems concepts and applications
- LO2: Apply image processing
- LO3: Implement feature extraction
- LO4: Apply machine learning based computer vision methods
- LO5: Implement video analytics algorithms
- LO6: Evaluate edge vs cloud-based computer vision systems

ELO	Statement	Abilities	Knowledge	Delivery topic
ELO1	Understand basic vision systems concepts and applications	A1	K1, K2	Topic 01
ELO2	Apply image processing	A2	K3, K4	Topic 02
ELO3	Implement feature extraction	A3, A4	K5, K6, K7	Topic 03
ELO4	Apply machine learning based computer vision methods	A5	K8, K9	Topic 04
ELO5	Implement video analytics algorithms	A6	K10	Topic 04
ELO6	Evaluate edge vs cloud-based computer vision systems	A7, A8	K11, K12, K13	Topic 05

Assessment

- Written Assessment WA(Q&A) — 13 open-ended short-answer questions covering K1 to K13, 60 minutes, open book, assessor-to-candidate ratio 1:3 to 1:10.
- Practical Performance (PP) — 5 scenario-based tasks covering A1 to A8, 90 minutes, open book, assessor-to-candidate ratio 1:3 to 1:10.
- Total assessment time: 2.5 hours (150 minutes) on Day 2.
- Format: Open Book — course slides, Learner Guide and approved course materials only.
- Both instruments are conducted on Day 2 from 16:10, after the Briefing for Assessment and the Assessment digital attendance.
- An oral clarification of up to 5 minutes may be conducted for each instrument to close minor performance gaps; that time sits outside the assessment duration.
- A minimum of 75% attendance on the SSG digital attendance record is required, and candidates must be assessed as 'Competent' in every ability and knowledge statement to be eligible for WSQ funding.

Instrument	Item	Criterion	Practised in
WA(Q&A)	Question 1	K1	Lab 01
	Question 2	K2	Lab 01
	Question 3	K3	Lab 02, Lab 03, Lab 04
	Question 4	K4	Lab 03, Lab 04, Lab 05
	Question 5	K5	Lab 06
	Question 6	K6	Lab 06
	Question 7	K7	Lab 07
	Question 8	K8	Lab 08

Instrument	Item	Criterion	Practised in
	Question 9	K9	Lab 08
	Question 10	K10	Lab 09, Lab 10, Lab 11
	Question 11	K11	Lab 12
	Question 12	K12	Lab 12
	Question 13	K13	Lab 12
PP	Task 1	A1	Lab 01
	Task 2	A2, A3	Lab 02, Lab 03, Lab 04, Lab 05, Lab 07
	Task 3	A4	Lab 06, Lab 07
	Task 4	A5, A6	Lab 08, Lab 09, Lab 10, Lab 11
	Task 5	A7, A8	Lab 12

Course Schedule

Slide references below are generated from the built deck (Generative AI for Image and Video Creation-v11.0.pptx, 204 slides) and are exact for this version. Any change to the deck regenerates this table.

Day 1 — Foundations, image generation and editing, and visual feature analysis

Time	Min	Component	Session	Slides
08:45–09:00	15	ADMIN	Course welcome, attendance, introductions and LMS access; outside programme minutes	—
09:00–10:15	75	CR	Topic1 / LU1: vision and generative pipelines; facilitated Lab01 needs-analysis discussion (K1,K2; A1)	Slides 24–54
10:15–10:30	15	BREAK	Tea break; excluded	—
10:30–11:45	75	CR	Topic2 / LU2: image representation, filtering, geometric transforms and controlled generation/editing (K3,K4)	Slides 55–94
11:45–12:45	60	PP	LU2 practical: Labs02,03,04,05 core exercises, 15 minutes each; extensions completed after class (A2)	Slides 90, 91, 92, 93
12:45–13:45	60	LUNCH	Lunch; excluded	—
13:45–15:00	75	CR	Topic3 / LU3: local/global descriptors, visual style and consistency (K5,K6,K7)	Slides 95–120
15:00–15:15	15	BREAK	Tea break; excluded	—
15:15–16:15	60	PP	LU3 practical: Labs06 and07, 30 minutes each (A3,A4)	Slides 118, 119
16:15–17:30	75	CR	Topic4 / LU4: learned vision models, segmentation/detection/recognition and evaluation (K8,K9)	Slides 121–161
17:30–18:30	60	PP	LU4 practical part1: Lab08 object detection/evaluation (A5)	Slides 157

Day 1 programme time: 480 minutes (8 hours). Administration, tea breaks and lunch are additional and are shown for the trainer's timetable only.

Day 2 — Video creation and analytics, cloud-versus-edge evaluation, and assessment

Time	Min	Component	Session	Slides
08:45–09:00	15	ADMIN	Attendance and technical setup; outside programme minutes	—
09:00–10:00	60	PP	LU4 practical part2: Labs09 storyboard/video prompts and11 captions/export, 30 minutes each (A5,A6)	Slides 158, 160
10:00–10:15	15	BREAK	Tea break; excluded	—
10:15–11:30	75	CR	Topic4 / LU5: temporal consistency, tracking, generative video and event analysis (K10)	Slides 121–161
11:30–12:30	60	PP	LU5 practical: Lab10 motion, tracking and temporal continuity (A6)	Slides 159
12:30–13:30	60	LUNCH	Lunch; excluded	—

Time	Min	Component	Session	Slides
13:30–14:45	75	CR	Topic5 / LU6: architectures, protocols and measured deployment trade-offs (K11,K12,K13)	Slides 162–194
14:45–15:00	15	BREAK	Tea break; excluded	—
15:00–16:00	60	PP	LU6 practical: Lab12 cloud-edge architecture and evaluation (A7,A8)	Slides 190
16:00–16:10	10	ADMIN	Assessment briefing and attendance; outside programme minutes	—
16:10–17:10	60	AS	Written Assessment: 13 open-ended questions, K1-K13, individual/open book	Slides 201–203
17:10–18:40	90	AS	Practical Performance: five tasks, A1-A8, individual/open book; submission and sign-off	Slides 201–203

Day 2 programme time: 480 minutes (8 hours). Administration, tea breaks and lunch are additional and are shown for the trainer's timetable only.

Topic Detail and Slide Ranges

Topic	Learning Unit / ELO	K / A	Slides	CR	PP
01 Generative AI Fundamentals for Image and Video Creation	LU1 · ELO1	K1, K2, A1	24–54	75	0
02 AI Image Generation, Editing and Visual Enhancement	LU2 · ELO2	K3, K4, A2	55–94	75	60
03 AI-Based Visual Features, Styles and Consistency	LU3 · ELO3	K5, K6, K7, A3, A4	95–120	75	60
04 Generative AI Video Creation, Animation and Editing	LU4 + LU5 · ELO4 + ELO5	K8, K9, K10, A5, A6	121–161	150	180
05 Evaluating Cloud and Edge AI Creative Workflows	LU6 · ELO6	K11, K12, K13, A7, A8	162–194	75	60

Hands-On Lab Reference

Every lab folder is independently usable: it carries its own numbered steps, mock data, local reference media, prompt pack, expected outputs, evidence checklist and troubleshooting table, and requires no paid service. Each learner-facing Markdown file has a same-basename PDF counterpart in the same folder.

Lab	Title	Topic	A	K	Min	Slide	Folder
01	Vision-System Needs Analysis and Generative Pipeline Specification	01	A1	K1, K2	in LU1	53	labs/lab-01-vision-needs-analysis/
02	Image and Video Data Representation	02	A2	K3	15	90	labs/lab-02-image-video-representation/
03	Prompt Experiment Matrix for Controlled Image Generation	02	A2	K3, K4	15	91	labs/lab-03-prompt-experiment-matrix/
04	Masking, Inpainting, Outpainting and Background Swap	02	A2	K3, K4	15	92	labs/lab-04-masking-inpainting-outpainting/
05	Filtering, Enhancement and Restoration Measured with PSNR and SSIM	02	A2	K4	15	93	labs/lab-05-filtering-enhancement-metrics/
06	Local Features: Colour Segmentation, Canny Edges and Corner Keypoints	03	A4	K5, K6	30	118	labs/lab-06-local-features-edges-keypoints/
07	Global Descriptors, Template Matching and Brand-Consistency Scoring	03	A3, A4	K7	30	119	labs/lab-07-global-descriptors-consistency/
08	Object Detection and Quantitative Evaluation	04	A5	K8, K9	60	157	labs/lab-08-object-detection-evaluation/

Lab	Title	Topic	A	K	Min	Slide	Folder
09	Storyboard to Video Prompt Pack and Deterministic Animatic	04	A6	K10	30	158	labs/lab-09-storyboard-video-prompting/
10	Motion, Tracking and Temporal Continuity Measurement	04	A6	K10	60	159	labs/lab-10-motion-tracking-continuity/
11	Captions, Overlays, Safe Areas and Export Profiles	04	A6	K10	30	160	labs/lab-11-captions-overlays-export/
12	Cloud-Edge Architecture Design and Measured Trade-offs	05	A7, A8	K11, K12, K13	60	190	labs/lab-12-cloud-edge-architecture-tradeoffs/

Instructional and Assessment Methods

Method	Ratio	Justification
Lecture	1:3 to 1:10	Transfers the underpinning knowledge for K1–K13 from facilitator to learners.
Didactic questioning	1:3 to 1:10	Confirms learners are following the explanation and can restate the mechanism, not just the term.
Demonstration	1:3 to 1:10	The facilitator runs each lab script once so learners see the expected output before attempting it.
Hands-on practical	1:3 to 1:10	Elicits performance for A1–A8 and lets the facilitator give feedback on the gaps each learner actually has.
WA(Q&A), 60 min	1:3 to 1:10	Direct written evidence that the learner holds the underpinning knowledge K1–K13.
Practical Performance, 90 min	1:3 to 1:10	Direct product evidence that the learner can perform A1–A8 on a scenario-based problem.

Resources Required

- One laptop per learner (Windows or macOS) with Python 3.10 or later, opencv-python 4.10 or later and NumPy installed before the class.
- The labs/ folder, distributed from the LMS/TMS portal. All reference media is included locally; nothing is downloaded during the class.
- Projector and whiteboard.
- Slide deck: Generative AI for Image and Video Creation-v11.0.pptx (204 slides) and its PDF.
- Learner Guide: LG-Generative AI for Image and Video Creation.docx and its PDF and Markdown mirror.
- Assessment papers and answer keys (trainer copy only, not distributed).
- Internet access is OPTIONAL. It is used only for the optional free-tier generation steps in Labs 03, 04 and 09; every lab completes without it.

Source Register

Every external claim in the slides and the Learner Guide traces to one of the following. Vendor documentation was verified on 6 September 2026; model identifiers change, and the register records which were deprecated at that date.

Ref	Source	Locator
S1	Course Proposal Tertiary Infotech CA-WSQ-2020-013290-v2 (SSG-approved)	Part 4 Section E, Curriculum Key Features table

Ref	Source	Locator
S2	Assessment Plan_OpenCV_TGS-2020505925_V1.0 (28 Dec 2020)	Assessment Duration table; WA and PP specification tables
S3	Competency Mapping.docx (TSC ICT-DIT-4022-1.1)	Course Mapping section and ELO/A-code matrix
S4	Computer Vision Technology.docx (Skills Framework TSC extract)	TSC Proficiency Level 4 (ICT-DIT-4022-1.1) knowledge and abilities columns
S5	Tertiary Courses public registration page	https://www.tertiarycourses.com.sg/ws-q-generative-ai-for-image-and-video-creation.html
S6	M. Tschochohei and F. Schenker, 'Multimodal Generative AI in the Enterprise: From Pixels to Profit', Packt, June 2026, ISBN 978-1-80611-167-1	Ch.5 pp.56-62
S7	Tschochohei and Schenker (2026), as above	Ch.5 pp.65-67
S8	Tschochohei and Schenker (2026), as above	Ch.5 pp.68-75
S9	Tschochohei and Schenker (2026), as above	Ch.5 pp.76-77 and pp.82-83
S10	Tschochohei and Schenker (2026), as above	Ch.7 pp.106-109
S11	Tschochohei and Schenker (2026), as above	Ch.7 pp.110-119
S12	Tschochohei and Schenker (2026), as above	Ch.8 pp.127-132 and pp.140-143
S13	Tschochohei and Schenker (2026), as above	Ch.9 pp.150-152
S14	A. Mewada, M. A. Ansari, S. Ahmad and N. Singh (eds), 'AI-Generated Image and Video Synthesis', IEEE Press / Wiley, 2026, ISBN 9781394403110	Ch.1 pp.1-11
S15	Mewada et al. (2026), as above	Ch.3 pp.35-44 (Vishnoi, Sisodia, Khare and Yadav, 'Sensors-to-synthesis: Edge AI and IoT for Generative Visual Systems')
S16	Mewada et al. (2026), as above	Ch.4 pp.47-54 (Sibi, Pantola and Gupta, 'Detecting AI-generated Images in the Social Media Era: A Deep Learning Approach with GenReal Dataset')
S17	Mewada et al. (2026), as above	Ch.7 pp.94-98 (Dubey, Pinheiro, Singh, Ansari and Kumar, 'Advanced Foundations and Future Trends in Generative AI for Visual Media')
S18	Mewada et al. (2026), as above	Ch.7 pp.98-100
S19	Mewada et al. (2026), as above	Ch.11 pp.147-160 (Dewang, Mewada, Omkar, Jaiswal and Singh, 'Hybrid Neural Networks for Robust Deepfake Detection')
S20	Google AI for Developers — Gemini API image generation documentation	https://ai.google.dev/gemini-api/docs/image-generation (verified 6 Sep 2026)
S21	Google AI for Developers — Gemini API Imagen documentation	https://ai.google.dev/gemini-api/docs/imagen (verified 6 Sep 2026)
S22	Google AI for Developers — Veo video generation documentation	https://ai.google.dev/gemini-api/docs/veo (verified 6 Sep 2026)
S23	OpenCV 4.13 Python package, verified locally on the delivery image	cv2 4.13.0 (opencv-python 4.13.0.92)